



National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

Sir Nasir Mehmood

Submitted by:

Eman Marium Tariq Rao

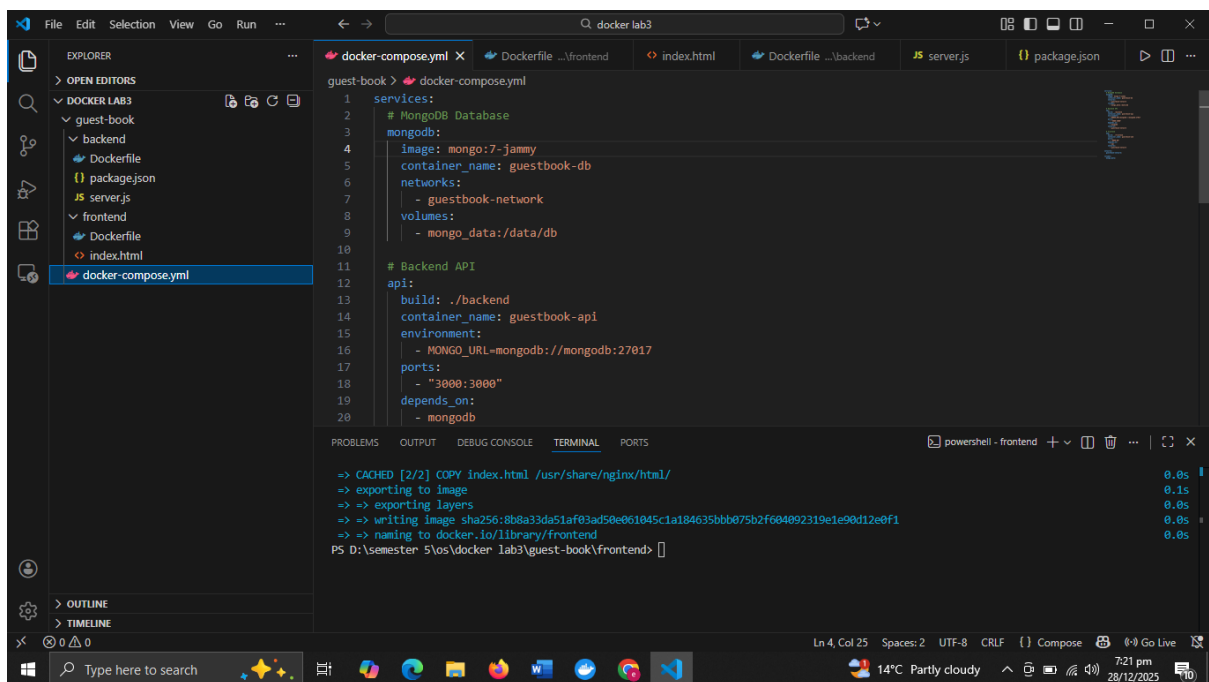
Reg number:

23-NTU-CS-1150

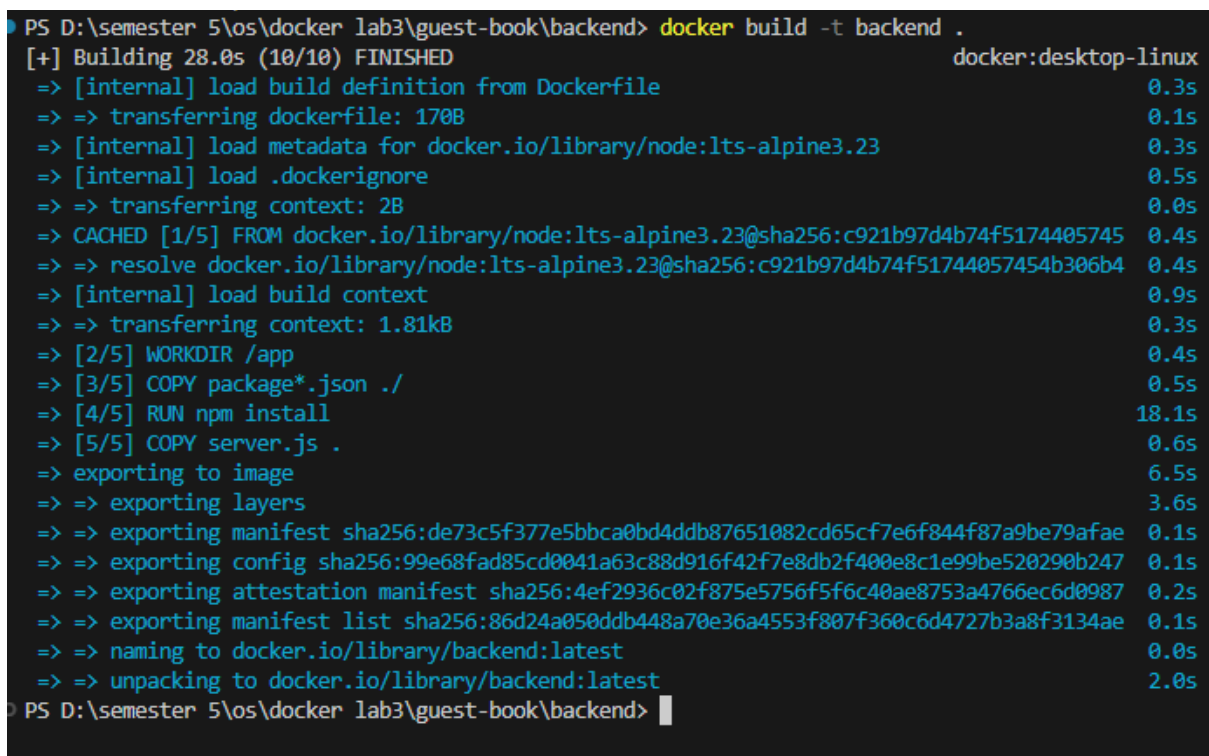
Semester:

05

Screenshot of project structure:



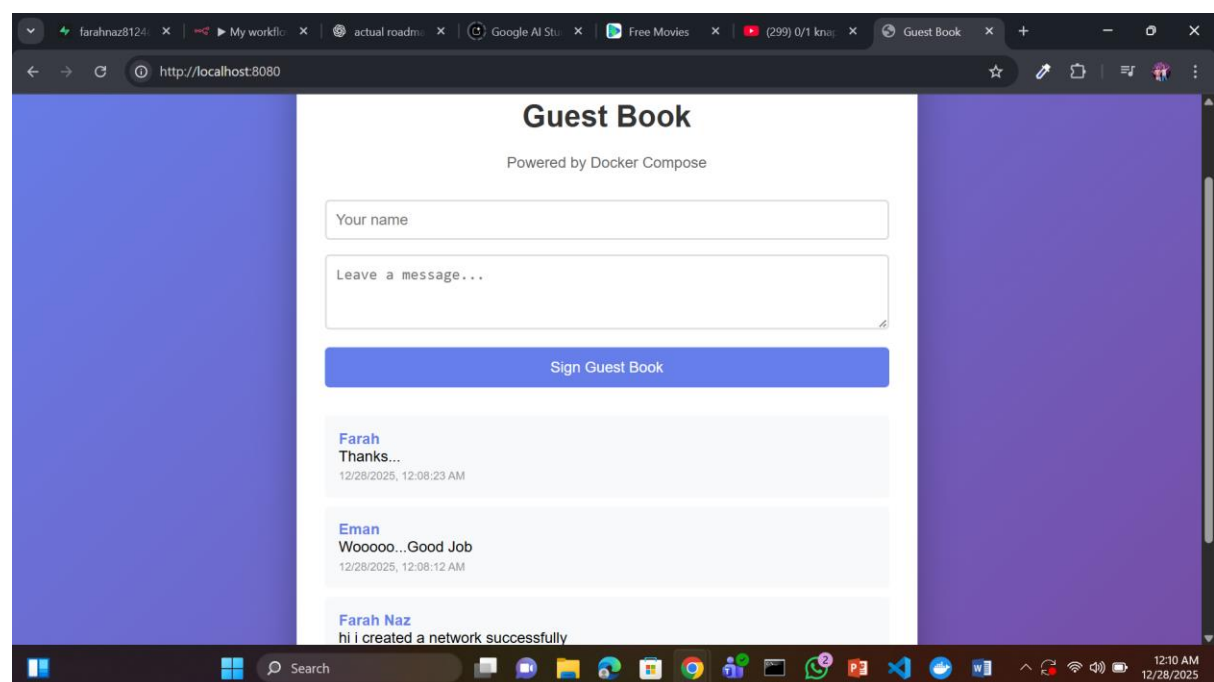
Output of building command “docker build -t backend .”



Output of building command “docker build -t frontend .”

```
PS D:\semester 5\os\docker lab3\guest-book\frontend> docker build -t frontend .
[+] Building 3.1s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 105B
=> [internal] load metadata for docker.io/library/nginx:trixie
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 32B
=> [1/2] FROM docker.io/library/nginx:trixie@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
=> CACHED [2/2] COPY index.html /usr/share/nginx/html/
=> exporting to image
=> => exporting layers
=> => writing image sha256:8b8a33da51af03ad50e061045c1a184635bbb075b2f604092319e1e90d12e0f1
=> => naming to docker.io/library/frontend
PS D:\semester 5\os\docker lab3\guest-book\frontend>
```

Webpage at localhost:8080:



- Copy contents of docker-compose and both Dockerfiles in the pdf.

Docker compose file:

```
services:
  # MongoDB Database
  mongodb:
    image: mongo:7-jammy
    container_name: guestbook-db
    networks:
      - guestbook-network
    volumes:
      - mongo_data:/data/db
```

```

# Backend API
api:
  build: ./backend
  # image: <image_name>
  container_name: guestbook-api
  environment:
    - MONGO_URL=mongodb://mongodb:27017
  ports:
    - "3000:3000"
  depends_on:
    - mongodb
  networks:
    - guestbook-network
# Frontend
web:
  build: ./frontend
  # image: <image_name>
  container_name: guestbook-web
  ports:
    - "8080:80"
  depends_on:
    - api
  networks:
    - guestbook-network
networks:
  guestbook-network:
volumes:
  mongo_data:

```

Dockerfile of backend:

```

FROM node:lts-alpine3.23
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY server.js .
EXPOSE 3000
CMD ["npm", "start"]

```

Dockerfile of frontend:

```

FROM nginx:trixie
COPY index.html /usr/share/nginx/html/
EXPOSE 80

```